Native bracket steady thermal study

Select a native PicoGK bracket or its structural study and press Analyze bracket temperature. This uses the selected registered STL, including holes and root transition, and registers a separate derived run. No mechanical result is overwritten. The fixed supported geometry is 60 × 30 × 40 mm with 3.5 mm plates and mounting holes.

The existing isolated CalculiX/Gmsh worker builds C3D10 meshes at 2.5, 1.8 and 1.5 mm and solves steady heat conduction with thermal degree of freedom 11. Geometry is in mm, heat in W and temperature in °C. Constant illustrative conductivity is 0.015 W/(mm·K), equivalent to 15 W/(m·K). It is not a claimed material measurement. Rear upright nodes with $|x| < 0.05$ mm and $z > 3.5$ mm are fixed at 20 °C; free-end nodes with $|x-60| < 0.05$ mm and $z \leq 3.55$ mm are fixed at 80 °C. Other surfaces have natural insulated boundaries. There is no internal heat generation.

CalculiX RFL reactions are summed at each reservoir. Positive hot reaction is heat supplied; negative cold reaction is heat removed. Their sum should be zero. Volume-mean temperature uses exact integrated quadratic shape weights for straight-sided tetrahedra: -1/20 per corner and 1/5 per midside node, weighted by corner tetrahedron volume. Full nodal temperatures and reactions are saved in CSV. Nodal values are read from solver output; no synthetic temperature gradient is painted onto the STL.

Checks require hot/cold relative heat imbalance below 1e-6 on every mesh; prescribed temperature error below 1e-6 °C; all nodal temperatures within 19.9-80.1 °C; last total heat change below 5%; last volume-mean temperature change below 0.5 °C; and fine-grid volume agreement with the source STL within 2%. These are demonstration gates, not general engineering acceptance criteria. Local heat flux convergence is not established by total heat and mean-temperature convergence.

Before each bracket solve, a separate structured C3D20 bar verifies the thermal procedure: 60 × 10 × 3.5 mm, identical conductivity, insulated sides, 20/80 °C ends. Its exact temperature is T=20+x_mm and heat flow is 0.525 W. Maximum temperature error and relative heat-flow error must each be below 1e-5. This verifies the CalculiX thermal procedure using a different element family; it is not a complete validation of the STL-derived C3D10 model. Bar inputs, outputs and check results are preserved in the bracket archive.

Saved artifacts include source STL and geometry metadata, thermal requirements/results, all three input decks, native Gmsh meshes/logs, CalculiX outputs/logs, nodal CSVs, fine boundary preview, report and manifest. The source parent manifest, STL, adapter, meshing source and solver binary are fingerprinted. Failed solver diagnostics are retained separately. Temperature preview requests verify the entire saved run before reading the archived fine surface; the response identifies its source archive hash. The browser checks that identity against the selected run.

Enable Solver colors to display saved fine-mesh nodal temperatures on the quadratic boundary, subdivided into four linear display triangles per face. Blue is the cold end of the numerical range and red the hot end. There is no deformation. This is a temperature field, not stress or heat flux. The original solid is available by disabling colors. The displayed 20-80 °C nodal range is imposed by ideal reservoirs; it does not represent a demonstrated service temperature range.

The study status is thermal_numerical_checks_passed_not_engineering_validated or thermal_numerical_checks_failed. Ideal reservoirs, illustrative conductivity and insulated surfaces do not establish realistic mounting/contact, convection, radiation, heat generation, transient response, temperature-dependent material behavior or thermal expansion. No thermal/mechanical coupling or design acceptance is granted.

CalculiX thermal keywords follow the author's [CalculiX manual](https://www.dhondt.de/ccx_2.11.pdf); the running pinned solver is version 2.20 and its input/output behavior is tested directly here.